CSM
Spring 2026

EECS 16B
Worksheet 5

1. Phasor Analysis

Any sinusoidal time-varying function $v(t)$, representing a voltage or a current, can be expressed in the form

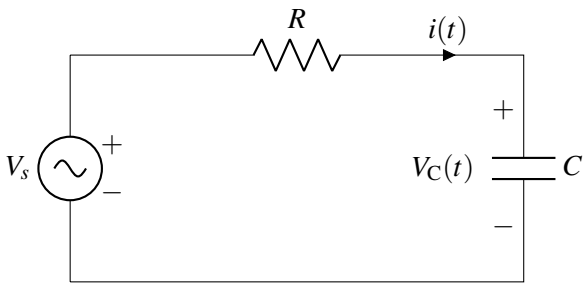
$$v(t) = \widetilde{V}e^{j\omega t} + \overline{\widetilde{V}}e^{-j\omega t}, \tag{1}$$

For a sinusoidal cosine wave $v(t) = V_0\cos(\omega t + \phi)$, a phasor is defined as:

$$\widetilde{V} = \frac{V_0}{2}e^{j\phi} \text{ and its complex conjugate } \overline{\widetilde{V}} = \frac{V_0}{2}e^{-j\phi}$$

$\widetilde{V}$ is a time-independent function, and is referred to as the phasor representation of $v(t)$.

The phasor analysis method consists of five steps. Consider the RC circuit below.



The voltage source is given by

$$V_s(t) = 8\cos\left(\omega t - \frac{\pi}{4}\right), \tag{2}$$

with $\omega = 10^3$ rad/s, $R = 1$ kΩ, and $C = 1$ μF.

Our goal is to obtain a solution for $V_C(t)$ and $i(t)$ with the sinusoidal voltage source $V_s(t)$.

(a) **Step 1: Convert $v(t)$ as a sum of complex exponentials**

All voltages and currents with known sinusoidal functions should be expressed in the standard exponential format. **Convert $v_s(t)$ into an exponential and write down its phasor representation $\widetilde{V}_s$.**

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(b) **Step 2: Transform circuits to phasor domain**

The voltage source is represented by its phasor $\widetilde{V}_s$. The current $i(t)$ is related to its phasor counterpart $\widetilde{I}$. What are the phasor representations of R and C?

(c) **Step 3: Cast KCL and/or KVL equations in phasor domain**

Use Kirchhoff's laws to write down a loop equation that relates all phasors in Step 2.

(d) **Step 4: Solve for unknown variables**

Solve the equation you derived in Step 3 for $\widetilde{I}$ and $\widetilde{V}_c$. What is the polar form of $\widetilde{I}$ ($Ae^{i\theta}$, where A is a positive real number)?

(e) **Step 5: Transform solutions back to time domain**

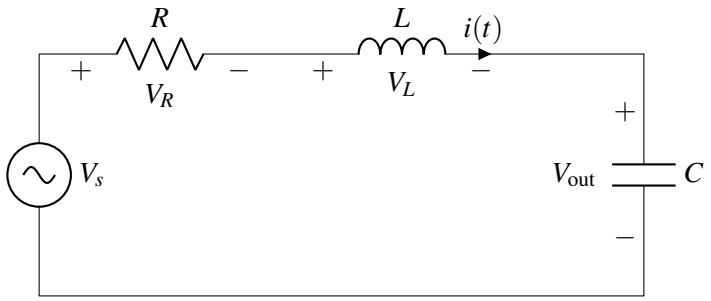
To return to time domain, we apply the fundamental relation between a sinusoidal function and its phasor counterpart. What is $i(t)$ and $v_c(t)$? What is the phase difference between $i(t)$ and $v_c(t)$?

It's important to keep in mind that the phasor analysis above only applies to sinusoidal inputs. Don't make the mistake of doing phasor analysis on DC inputs!

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2. RLC circuit Phasor Analysis

In this question, we will take a look at an electrical systems described by second order differential equations and analyze it using the phasor domain. Consider the circuit below where $R = 3\text{k}\Omega$, $L = 1\text{mH}$, $C = 100\text{nF}$, and $V_\text{s} = 5\cos\left(1000t + \frac{\pi}{4}\right)$:



(a) What are the impedances of the resistor, inductor and capacitor, Z_R, Z_L, and Z_C?

(b) Solve for $\widetilde{V}_\text{out}$ in phasor form.

(c) What is V_out in the time domain?

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(d) Solve for the current $i(t)$

(e) Solve for the transfer function $H(\omega) = \frac{\widetilde{V}_{\text{out}}}{\widetilde{V}_{\text{s}}}$
Leave your answer in terms of R, L, C, and ω.

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3. RC Circuit

Consider the circuit below, assume that when $t \leq 0$, the capacitor has no charge stored ($V_c(t=0)=0$). At $t=0$, the switch closes. Assume that $V_s = 5\,\text{V}$, $R = 100\,\Omega$, and $C = 10\,\mu\text{F}$.

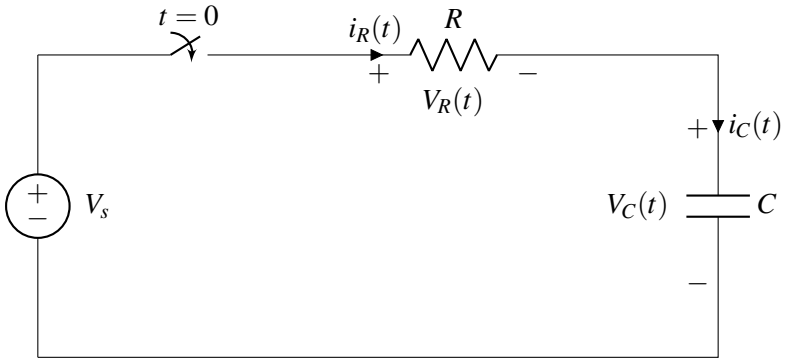


Figure 1: RC Circuit with Voltage Source

(a) Write out the KCL equations associated with the circuit when the switch is closed.

(b) Write out the differential equation for $V_c(t)$ after the switch closes.

(c) What is the initial condition for $V_c(t)$ (i.e. $V_c(t=0)$) and what is $V_c(t \to \infty)$?

(d) Using the initial conditions found in the previous parts, find an expression for $V_c(t)$ in terms of V_s, R, and C.

Aside: τ is the time constant that we denote for RC circuits. Its definition is the time required to charge the capacitor from an initial voltage of zero to about 63.2% of the value of V_s, or to discharge the capacitor to about 36.8% of its initial charge voltage. For an RC circuit it is equal to $\frac{1}{\alpha}$ where α is the eigenvalue of the RC circuit.

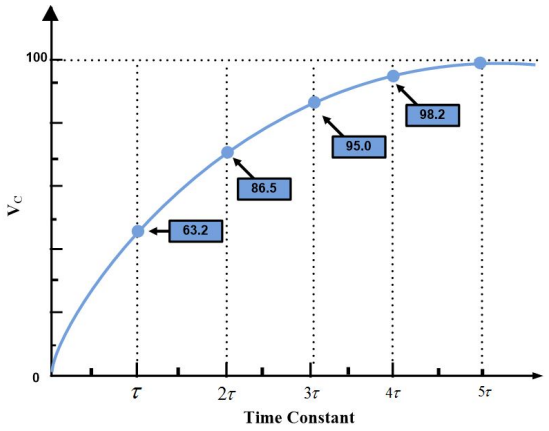


Figure 2: Different values of capacitor voltage at different times, relative to τ.

(e) OPTIONAL: On what order of magnitude of time (nanoseconds, milliseconds, 10's of seconds, etc.) does this circuit settle (V_c is $> 95\%$ of its value as $t \to \infty$)?

(f) OPTIONAL: Give 2 ways to reduce the settling time of the circuit if we are allowed to change one component in the circuit.

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